

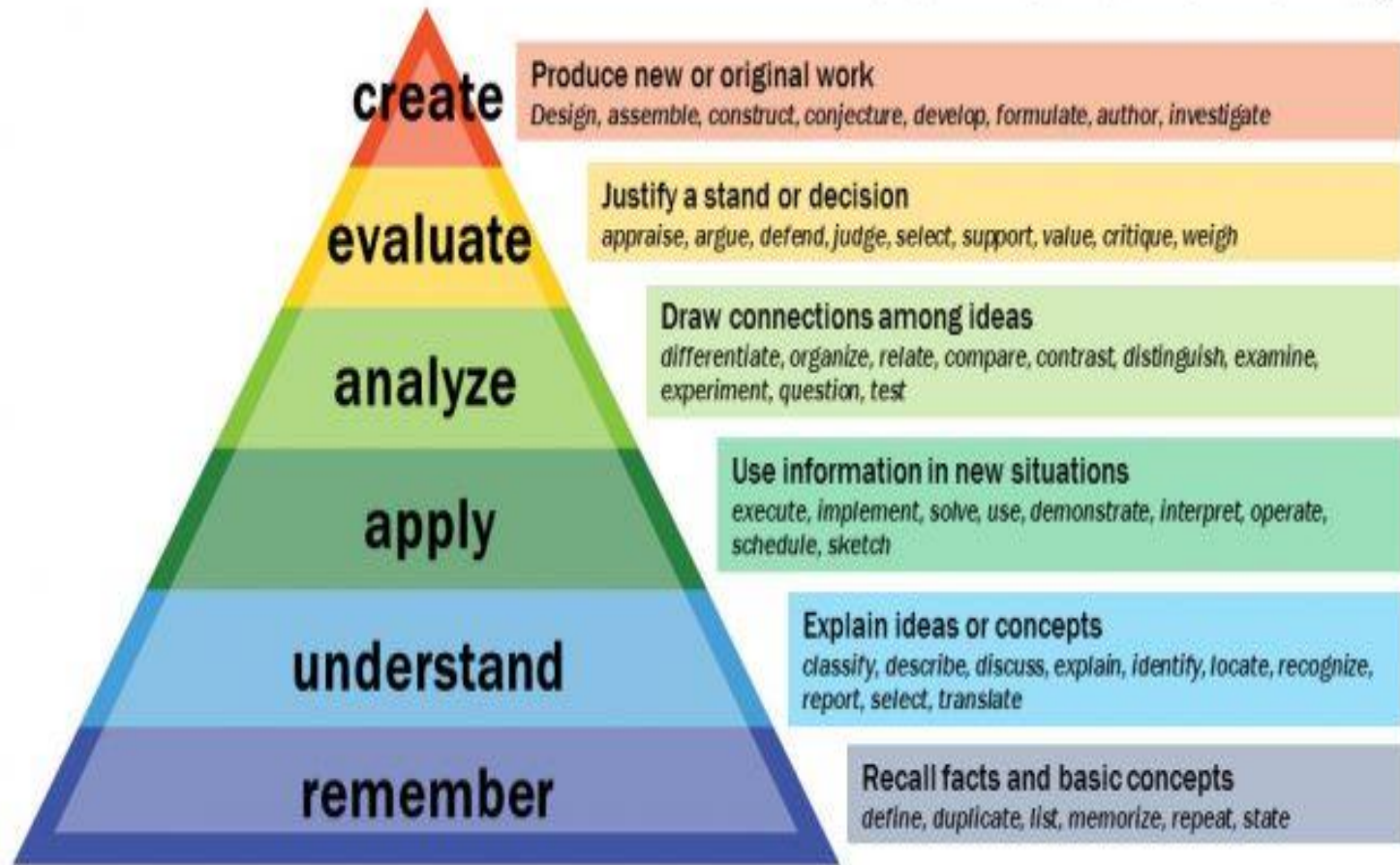
AY 2023-2024

Software Project Management

Course Objectives:

- ❖ To learn process of Software Project Management.
- ❖ To Study role of Project Manager in Project Management.
- ❖ To learn Agile Project Management Framework.
- ❖ To study various role of Agile Team and Tools.
- ❖ To understand project planning and tracking.

Bloom's Taxonomy



AY 2023-2024

SPM

- *Course Outcomes:*

- Student will be able to
- CO1: Understand the process of Software Project Management Framework and Apply estimation techniques. (Apply)
- CO2: Learn the philosophy, principles and lifecycle of an agile project. (Understand)
- CO3: Demonstrate Agile Teams and Tools and Apply agile project constraints and trade-offs for estimating project size and schedule (Apply)
- CO4: Explain Project Tracking and Interpretation of Progress Report (Understand)
- CO5: Analyze Problem statement and evaluate User Stories (Analyze)

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Unit 1 :

Linear Project Management Framework

- **Project Management Overview**
- **Software Project Life Cycle**
- **Project Management Process**
- **Team Structure**
- **Role of Project Manager**
- **Risk Management**

Need of Project Management

- In the late nineties, **Research studies carried** out by research groups :
 - Defense Science Board 1994, Standish Group 1995
 - Jone Group 1996, 2010

- Association for Information Systems 2007
- **The findings were :**
 - Software sometimes does not meet requirement.
 - Costs more than budgeted.
 - Takes longer time to complete than it is planned.
 - Maintenance is difficult.
 - Lack of communication,

So planning of projects has always been one of the major tasks of managers.
- However the success of a project will always rest on the abilities of a project manager and team members.

Overview of Project Management

- Project :

A Project is a *sequence of unique, complex and connected activities* having one goal or purpose and that must be completed by a *specific time within a budget* and according to specifications.



Overview of Project Management

- The Software project management :

It deals with properly planning and controlling the *software engineering activities to meet* software project goals for cost, schedule and quality.

- Project management process mainly consists of the *set of activities executed by* a project manager.
- It begins *before any technical activities* is initiated and continue throughout the development of computer software.

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- The objective of software development is to produce software which is
 - **easy to use**
 - **developed within cost and time constraints**
 - **Quality product**

UOP Q.

- What do you mean by Project ? How software projects are unique in their characteristics?

A **Project** is a sequence of unique, complex and connected activities having one goal or purpose and that must be completed by a specific time within a budget and according to specifications.

- A **software project** is developed for a customer according to his/her specification.
- The **Project Management** can be defined as the activity of defining, planning, directing, monitoring and controlling a project to develop an acceptable system within the allotted time and budget as per quality.

Software Project Characteristics (SPPU Q)

- Project has customers.
- It has definable inputs and end product.
- It is unique.
- It is temporary in nature and evolves as it progresses.
- It involves risk.
- It is process of working to achieve goals.
- It requires resources.
- It is associated with cost, schedule and quality.

Software Project Characteristics

- **Maintainability:**

Software must evolve to meet changing needs.

- **Dependability:**

Software must be reliable.

- **Efficiency :**

Software should not waste system resources.

- **Usability:**

It should be usable by the users for whom it is designed.

1.1 Overview of project Management



SPLC : Project Management Life Cycle

- Life Cycle :
The entire life of a project encompassing its conception, design, development, construction, operation, repair, and maintenance.
- PMLC Phases:
 - Phase1 : Project Initiation
 - Phase 2 : Project Planning
 - Phase 3 : Project Execution
 - Phase 4 : Project Closing
- Figure



Initiation Phase

1 The project need is identified in the initiation phase. It signals the start of a project.



Implementation phase

3 The project plan is put into action. This phase is when people are carrying out the tasks of their plan.



Planning phase

2 This phase is where the project solution is further explored and the steps necessary to achieve their goals.



Closure phase

4 This phase is the final closing of all project processes and handing over deliverables to the customer.



Phase I: Project Initiation

- Writing Project Initiation Document(PID) involves following:
- Purpose
- Project Scope
- Background
- Initial Project Plan
- Initial Quality Plan
- Initial Risk Plan
- Assumptions/Constraints.

Phase II: Project Planning :

- In project planning stage, the project manager reviews contractual commitments and creates a plan to meet them.
- Study of all issues including the following:
 - Define Objectives
 - Understand Constraints
 - Identify all subsystems & interfaces
 - Identify risks , costs & availability
 - Determine personnel needs
 - Establish schedule
 - planning for quality assurance, configuration management as well as risk management

Contents of Project Plan

1. Define the project objectives and scope.
2. Identify a suitable standard process for project execution.
3. Define a process for managing changes in requirements.
4. Estimate the efforts and Cost.
5. Plan for human resources and team organization.
6. Define the project milestones and create a schedule.
7. Define the quality objectives and a quality plan to achieve them.
8. Make a defect prevention plan.
9. Identify risks and make plans to mitigate them.
10. Define a training plan for the project.
11. Define project-tracking procedures.

Project Execution:

1. Execute the project as per the project plan.
2. Track the project status.
3. Review the project status with senior management.
4. Monitor compliance with the defined project process.
5. Analyze defects & reform defect prevention activities.
6. Monitor performance at the program level and risks.
7. Conduct milestone review & re plan it.

Project Closure:

1. This involves systematic windup of the project after customer acceptance.

2. Post –projects analysis constitutes the main activity. Process assets (i.e. materials such as- templates & guidelines used to aid in managing the process) are collected for future use and the lessons are recorded.
3. The main outputs of this phase are project closure report & the collected process assets.

Role of the Project Manager

-  Project Definition
-  Document Management
-  Stakeholder Engagement
-  Project Planning
-  Resource Allocation
-  Leadership and Motivation
-  Delivery and Control
-  Budget Management
-  Managing Dependencies
-  Attend Governance Meetings
-  Quality Assurance and Control
-  Change Control

Project Manager

-  Progress Monitoring
-  Risk & Issue Management
-  Project Closure
-  Review Lessons Learned

**Software Project
Management**

SPPU IMP Questions

- Define PMLC/SPLC
- Explain SPM Process in detail.
- Define role of Project Manager.